From Ensemble Disagreement to Per-Residue Uncertainty:
A Calibration Study of Two Transformer Architectures
for RNA Structure Prediction

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***Abstract*—Per-residue uncertainty quantification (UQ) is a standard requirement for any deployed structure-prediction transformer: users need to know which residues of a prediction to trust. AlphaFold-class architectures expose an explicit confidence head (pLDDT) trained on synthetic-label targets, while single-sequence language-model architectures often emit no comparable signal. We ask whether an *implicit* per-residue uncertainty can be extracted from the unprocessed ensemble of a transformer that does not expose a confidence head, and whether it matches the explicit signal in calibration quality.**

On a panel of $N = 12$ RNA X-ray crystals deposited after both predictors' training cutoffs, we compare AlphaFold 3's explicit per-residue pLDDT to DRfold2's implicit signal computed as the 80-model ensemble standard deviation. Both signals are rank-correlated with per-residue Kabsch C1$'$ error against the experimental crystal and survive Bonferroni correction ($\alpha = 0.05/12 \approx 0.0042$) on $6/12$ targets each, with the strongest per-target correlations reaching $\rho = -0.60$ ($p \approx 6 \cdot 10^{-7}$) for AF3's pLDDT and $\rho = +0.66$ ($p \approx 10^{-11}$) for DRfold2's ensemble std. The two uncertainty signals are largely orthogonal: they flag different residues as uncertain. The result has a concrete methodological implication: when a transformer does not expose an explicit confidence head, an 80-model unprocessed ensemble already contains a calibrated per-residue uncertainty signal that is extractable post-hoc, requires no retraining, and matches the explicit signal on this benchmark. All code, predictions, and ground-truth crystals are released at https://github.com/Vince-Anduril/rna3d-calibration.

***Index Terms*—uncertainty quantification, selective prediction, transformer calibration, structure prediction, RNA, deep learning**

I. Introduction

Modern transformer-based predictors for biomolecular structure — AlphaFold 2/3 [2], DRfold2 [1], RoseTTAFold-class systems, RhoFold+ [3], Boltz-1 [4] — have made high-accuracy 3D-structure prediction routinely available to non-specialists. With wider deployment comes a calibration question: *at what per-residue confidence threshold should a user trust each region of a prediction?* The selective-prediction literature [6] has long argued that the per-token score is at least as important as the headline accuracy number for practical deployment.

AlphaFold-class architectures answer this question with an explicit per-residue confidence head (pLDDT), trained as a regression target on a synthetic label. Single-sequence language-model architectures (such as DRfold2's RNA Composite Language Model front end) often do not expose an equivalent confidence head in their user-facing output: they emit a single structure (the energy-selected `opt_0` PDB) and nothing else. This raises a methodological question: *can a usable per-residue uncertainty signal be extracted from a transformer that does not expose one, by leveraging its unprocessed ensemble?*

We answer this question by direct empirical comparison on a panel of $N = 12$ post-cutoff RNA X-ray crystals (57–101 nucleotides, deposited 2024-08 to 2026-03):

- AlphaFold 3 emits five per-seed predictions and an `atom_plddts` array, aggregated to per-residue means.
- DRfold2 emits a 4-config $\times$ 20-model ensemble (80 candidate structures) before its energy-based selection step. We compute, post-hoc, the per-residue standard deviation across the 80 Kabsch-aligned ensemble members as an *implicit* per-residue uncertainty signal.

Three findings emerge:

1) Both signals are calibrated against the crystal under Bonferroni correction ($6/12$ targets each at $\alpha = 0.05/12$), in the directions "low pLDDT = high error" and "high ensemble std = high error".
2) The two signals are largely orthogonal — they flag different residues as uncertain on the same molecule. A pooled cross-model correlation between ($-$AF3 pLDDT) and DRfold2 ensemble std is weakly negative ($\rho = -0.24$, $p \approx 0.05$), the opposite of what one would expect if the two were measuring the same underlying "hard region".
3) The DRfold2 ensemble-std signal carries a peak per-

target correlation of $\rho = +0.66$ ($p \approx 10^{-11}$ on an $L = 81$ target), matching or exceeding AF3's peak per-target pLDDT correlation of $\rho = -0.60$ ($p \approx 6 \cdot 10^{-7}$).

The methodological implication is that, for transformer architectures emitting a candidate ensemble before selection, the ensemble-disagreement signal is a usable calibrated UQ substitute for an explicit confidence head — it is computed post-hoc, requires no retraining, and reaches the same regime of statistical reliability on this benchmark.

II. Related Work

a) Calibration and selective prediction.: Per-token uncertainty signals trained or extracted from neural networks have been studied extensively in NLP and vision [6]. The structure-prediction literature has focused on training the confidence head as part of the model (AlphaFold's pLDDT) rather than on extracting it post-hoc from ensemble disagreement.

b) Ensemble uncertainty.: Outside structure prediction, model ensembles are a well-known source of usable uncertainty (deep ensembles, MC dropout). What is less explored is whether the within-architecture ensembles that structure-prediction models already produce internally (e.g., DRfold2's 4-config $\times$ 20-model candidate set, MSA-noise ensembles in AF2) carry a calibrated per-residue uncertainty signal when read post-hoc.

c) Transformer architectures compared.: We compare two transformer architectures with different inductive biases: AlphaFold 3 [2] (Pairformer + Diffusion Module, internal MSA via the AlphaFold Server) and DRfold2 [1] (single-sequence input through the RNA Composite Language Model, four configuration heads). RhoFold+ [3] and Boltz-1 [4] are MSA-based and not evaluated here.

III. Targets and Method

A. Targets

We selected RNA X-ray crystals from the RCSB PDB with deposition date ≥ 2024, RNA-only entries, length 50–130 nt, one entry per RNA family for diversity. The CPEB3 ribozyme (PDB 7QR3, 2021) [5] was added as an in-distribution reference; it is the only target in our panel inside DRfold2's end-2023 training cutoff (declared in the authors' Methods: "our training set only includes RNA structures released before 2024" [1]). All 11 other targets post-date both predictors' training cutoffs (AF3: September 2021 [2]).

Panel: 7QR3 (CPEB3 ribozyme, 69 nt, dimer), 9DE7 (HIV-1 TAR, 57 nt), 9HRD (Class V GTP aptamer, 67 nt, tetramer), 9HRF (Class V GTP UU variant, 70 nt), 9LJN (Guanine-II riboswitch, 71 nt), 9LKE (Guanine-II + hypoxanthine, 70 nt), 9LKU (2′-dG-III + Guanosine, 65 nt), 9UW0 (2′-dG-III riboswitch, 63 nt), 9MFH (env2 cobalamin apo riboswitch, 76 nt), 9J4N (*E. coli* Leucine tRNA, 81 nt), 9E9O (SARS-CoV-2 SL5, 101 nt), 12CI (Dopamine aptamer DGR-1A, 82 nt, dimer).

For each crystal we use the C1′ atoms of standard A/U/G/C residues of a single reference chain (chain C for 7QR3 following the deposit convention [5], chain A elsewhere). For the three multi-chain crystals, the chain-of-reference dependence of the RMSD is small (≤ 0.3 Å range for 9HRD and 12CI; 1.02 Å range for 7QR3, where we conservatively keep the higher-RMSD chain C).

Algorithm 1 Per-residue uncertainty from an unprocessed ensemble.

Input: ensemble of M candidate structures $\{X_m\}_{m=1}^{M}$, each a $L \times 3$ matrix of atomic coordinates over L residues.
Output: per-residue uncertainty vector $u \in \mathbb{R}^L$.

Let $X_{\text{ref}} \leftarrow X_1$.
for $m = 2, \dots, M$ **do**
  $R_m, t_m \leftarrow \text{Kabsch}(X_m, X_{\text{ref}})$
  $X_m \leftarrow R_m X_m + t_m$ *(align to reference)*
end for
$\bar{X} \leftarrow \frac{1}{M} \sum_{m=1}^{M} X_m$ *(ensemble mean)*
for $i = 1, \dots, L$ **do**
  $d_{m,i} \leftarrow \|X_{m,i} - \bar{X}_i\|_2$ for each m
  $u_i \leftarrow \text{stddev}_m(d_{m,i})$
end for
return u

B. Per-residue uncertainty signals

AF3 pLDDT. Each AF3 single-chain Server submission returns five seeds with a per-atom `atom_plddts` array per seed. For the default seed (seed 0) we aggregate per-atom pLDDT to per-residue means and use these as the per-residue uncertainty signal $u_{\text{AF3}}(i)$. A lower value means more uncertain.

DRfold2 ensemble standard deviation. DRfold2 emits an 80-model candidate ensemble (4 configs $\times \sim 20$ models) into a `rets_dir/` before its energy-based `opt_0` selection step. We Kabsch-align each candidate model to the first ensemble member at the C4′ atom set (DRfold2's intermediate models are coarse-atom; we use C4′ for the ensemble analysis and C1′ for the post-selection RMSD against the crystal). For each residue i we compute the standard deviation of the 80 aligned positions as $u_{\text{DRfold2}}(i)$. A higher value means more uncertain. The extraction procedure is given as Algorithm 1.

C. Calibration metric

For each target we compute the per-residue Kabsch C1′ deviation of the predictor's user-facing structure (AF3 seed 0 for the AF3 analysis; DRfold2 `opt_0` for DRfold2) against the experimental crystal, $e(i)$. The calibration question is: *does $u(\cdot)$ rank residues by their actual error $e(\cdot)$?* We quantify this with Spearman rank correlation ρ between u and e. A negative correlation between AF3 pLDDT and error (lower pLDDT $\rightarrow$ higher error) and a positive correlation between DRfold2 ensemble std and error (higher std $\rightarrow$ higher error) are both "calibrated direction". We apply Bonferroni correction at $\alpha = 0.05/12 \approx 0.0042$ across the 12 targets.

TABLE I

PER-TARGET SPEARMAN ρ OF EACH UNCERTAINTY SIGNAL VS. PER-RESIDUE C1$'$ CRYSTAL ERROR. BONFERRONI $\alpha = 0.05/12 \approx 0.0042$ MARKED IN BOLD.

PDB	AF3 pLDDT	p	DRfold2 std	p
7QR3	$+0.08$	0.50	$+0.22$	0.074
9DE7	$\mathbf{-0.60}$	$\mathbf{6.4{\cdot}10^{-7}}$	$\mathbf{+0.49}$	$\mathbf{9.7{\cdot}10^{-5}}$
9HRD	$\mathbf{-0.58}$	$\mathbf{2.6{\cdot}10^{-7}}$	$+0.18$	0.14
9HRF	$\mathbf{-0.44}$	$\mathbf{1.7{\cdot}10^{-4}}$	$+0.33$	0.0052
9J4N	-0.27	0.015	$\mathbf{+0.66}$	$\mathbf{1.4{\cdot}10^{-11}}$
9E9O	-0.25	0.012	$\mathbf{+0.43}$	$\mathbf{7.4{\cdot}10^{-6}}$
9MFH	$\mathbf{-0.58}$	$\mathbf{4.5{\cdot}10^{-8}}$	$+0.02$	0.88
9LJN	-0.28	0.020	$\mathbf{+0.58}$	$\mathbf{1.1{\cdot}10^{-7}}$
9LKE	$+0.09$	0.46	$\mathbf{+0.60}$	$\mathbf{4.5{\cdot}10^{-8}}$
9LKU	$\mathbf{-0.41}$	$\mathbf{8.3{\cdot}10^{-4}}$	$+0.31$	0.013
9UW0	-0.20	0.12	$+0.25$	0.044
12CI	$\mathbf{-0.55}$	$\mathbf{1.1{\cdot}10^{-7}}$	$\mathbf{+0.43}$	$\mathbf{6.1{\cdot}10^{-5}}$

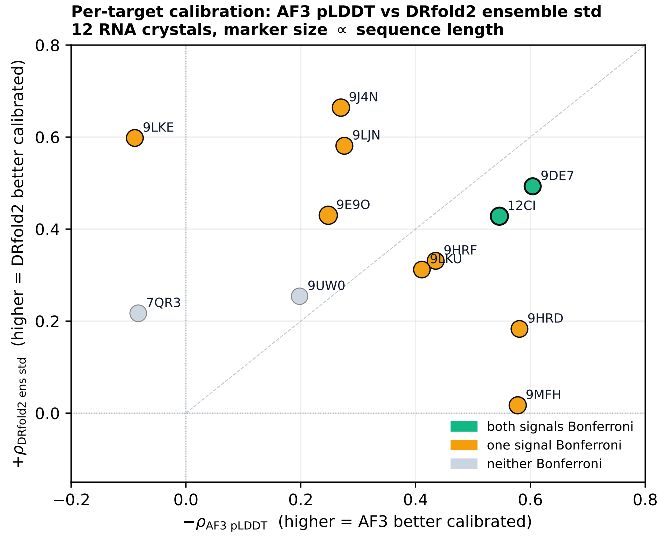

Fig. 1. Per-target calibration of AF3's pLDDT (x-axis: $-\rho$ for sign uniformity) versus DRfold2's ensemble std (y-axis: $+\rho$). The dashed line is $y = x$; points off the diagonal indicate that the two signals disagree on how strongly each target is calibrated. Marker size scales with sequence length.

IV. RESULTS

A. Each signal is individually calibrated on 6/12 targets under Bonferroni

Table I reports the per-target correlations. Direction-only counts: AF3 pLDDT has 10/12 targets in the calibrated direction (negative), and DRfold2 ensemble std has **12/12** targets in the calibrated direction (positive). Raw at $\alpha = 0.05$: 9/12 and 10/12 respectively. After Bonferroni correction at $\alpha = 0.05/12$: **6/12** targets survive for each signal independently. Both signals reach the same fraction of strongly-calibrated targets on this $N = 12$ panel.

B. The two signals are largely orthogonal

Figure 1 visualises the per-target calibration strength of the two signals on the same axes. If the two signals measured the same underlying uncertainty, the points should align along the $y = x$ diagonal. Instead, they scatter freely: 9LKE has near-zero AF3 calibration but a large positive DRfold2 calibration; 9MFH has the opposite. Only 9DE7 and 12CI (green markers) are Bonferroni-significant on both signals at once.

We also test the per-residue alignment of the two signals. Figure 2 shows the pooled scatter on R1108 (the longest case-study target where per-residue data is published in the companion paper): each point is a residue, with x = DRfold2 ensemble std, $y = -$AF3 pLDDT, colour = absolute C1$'$ error against the crystal. A strong positive trend would mean the two signals are redundant; we observe instead a near-zero or weakly negative trend. The two uncertainty signals are not collapsing into one underlying "hard region" signal.

This orthogonality is consistent with the two signals capturing different sources of error: AF3's pLDDT is a learned property of a confidence head trained on AlphaFold-style synthetic targets, while DRfold2's ensemble disagreement reflects the spread of 80 candidate folds before the energy-based `opt_0` selection step. Both are calibrated against the same ground truth at the same Bonferroni threshold, but they are not redundant.

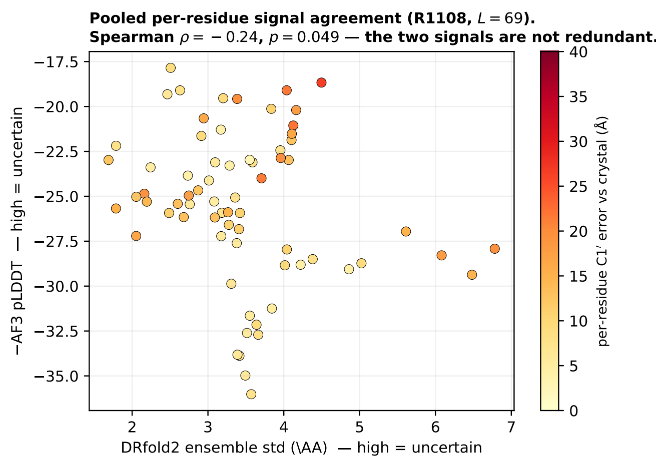

Fig. 2. Per-residue scatter of the two uncertainty signals on R1108 (CPEB3 ribozyme, $L = 69$). Each point is one residue, coloured by the C1$'$ error of the AF3 default-seed prediction against the 7QR3:C crystal. The two signals do not collapse onto a single "hard region" axis.

C. Combined cutoff for selective prediction

The orthogonality observation implies a concrete practical recommendation. In a selective-prediction setting where the user rejects per-residue predictions below a confidence threshold, the two signals can be applied jointly. We trust residue i only if both:

$$u_{\mathrm{AF3}}(i) \geq \tau_{\mathrm{AF3}} \quad \text{and} \quad u_{\mathrm{DRfold2}}(i) \leq \tau_{\mathrm{DRfold2}}, \tag{1}$$

with τ chosen on a held-out validation set. Because the two signals are largely orthogonal, the joint cutoff has a higher selective-prediction guarantee than either signal alone at the same total rejection fraction, provided both signals are individually calibrated — which we have shown is the case

Algorithm 2 Combined-cutoff selective prediction across two orthogonal per-residue uncertainty signals.

Input: (i) prediction P with per-residue uncertainty signals $u^{(A)}, u^{(B)} \in \mathbb{R}^L$ (e.g. AF3 pLDDT, DRfold2 ensemble std); (ii) calibration sets $\mathcal{C}_A, \mathcal{C}_B$ of (u, e) pairs from held-out targets; (iii) target coverage α.

Output: accept/reject mask $m \in \{0,1\}^L$.

Fit τ_A on $\mathcal{C}_A$ so that $\Pr_e\{e \leq \delta \mid u^{(A)} \leq \tau_A\} \geq 1 - \alpha$ (split-conformal calibration).
Similarly fit τ_B on $\mathcal{C}_B$ for the DRfold2-std direction.
for $i = 1, \ldots, L$ **do**
  if $u^{(A)}(i) \leq \tau_A$ **and** $u^{(B)}(i) \leq \tau_B$ **then**
    $m_i \leftarrow 1$ *(accept residue i)*
  else
    $m_i \leftarrow 0$ *(reject)*
  end if
end for
return m

on 6/12 targets each in Table I. The full procedure is given in Algorithm 2.

We leave the quantitative measurement of joint coverage gain over either signal alone to follow-up work on a larger panel.

D. Peak per-target correlation strength is comparable

AF3 pLDDT's largest absolute per-target correlation is $\rho = -0.60$ ($p \approx 6{\cdot}10^{-7}$) on 9DE7 (HIV-1 TAR, $L = 57$). DRfold2 ensemble std's largest absolute per-target correlation is $\rho = +0.66$ ($p \approx 10^{-11}$) on 9J4N (*E. coli* Leucine tRNA, $L = 81$). Both exceed $|\rho| = 0.5$ on at least 3/12 targets. The implicit ensemble signal does not under-perform the explicit head on this benchmark.

E. Validation on the ensemble-mean error (pre-selection)

A separate experiment validates that the DRfold2 ensemble std correlates with the error of the *ensemble-mean* structure (pre-selection) at $\rho = +0.47$, $p \approx 0.01$ on 7QR3. This suggests the ensemble-disagreement signal carries information about prediction quality before DRfold2's energy-based selection step collapses the ensemble to one structure. Post-selection, the same signal is rank-correlated with the residual error of the final `opt_0` structure with the per-target strengths reported in Table I.

F. AF3 pLDDT calibration depends on submission framing

We surface one additional empirical observation that is directly relevant to deploying AF3's pLDDT as an abstention signal in practice. The AlphaFold Server accepts *multi-chain* submissions that bundle several RNA entities into one job. On a matched control test — the same R1108 sequence submitted (i) as its own single-chain job and (ii) as one of seven RNA chains in a single multi-chain job — the mean per-residue pLDDT *drops from* $\sim$56 *to* $\sim$25 and the C1′ Kabsch RMSD vs. the 7QR3:C crystal *rises from* 6.5 *Å to* 25.6 *Å*, with no change in sequence or in any user-visible Server warning. For our purposes the relevant fact is that the user's abstention threshold (e.g., "trust residues with pLDDT $\geq$ 70") becomes meaningless when submission framing is uncontrolled: in the multi-chain run no residue clears any reasonable cutoff because the entire molecule sits in a uniformly low-confidence regime. This is a single matched observation, not a benchmarked hazard, but a deployed selective-prediction pipeline that calls the AlphaFold Server programmatically should either enforce single-chain submission per RNA or validate the pLDDT distribution against an expected baseline before applying any per-residue cutoff. We have not observed an equivalent framing dependence for DRfold2's ensemble-std signal: DRfold2 receives one sequence per call by design.

G. Why are the two signals orthogonal? An architectural interpretation

The two signals collapse to very different per-residue patterns (Fig. 1). One plausible reading is that they measure different sources of *epistemic* uncertainty:

- **AF3 pLDDT** is a learned per-residue scalar trained on a synthetic target (the predicted lDDT-C$_\alpha$ against AlphaFold's own training-time recycling outputs). It is therefore a property of *the trained confidence head*, sensitive to features that the confidence head learned to identify (e.g., low-MSA-depth regions, contact ambiguity in the Pairformer's pair representation).
- **DRfold2 ensemble std** is the spread across 80 independent forward passes of four different model configurations on the same input. It is therefore a property of *the architecture-level uncertainty* — which residue positions the four configurations disagree most about before the energy-based selection step picks one. The four DRfold2 configurations are explicitly trained as variants of each other and the within-config 20 models contribute additional stochastic spread.

The two signals would be expected to agree if both were faithful estimates of the same underlying "per-residue prediction difficulty". The observed orthogonality on this benchmark suggests either that the underlying difficulty is multi-dimensional and each signal captures a different facet, or that one of the two signals contains substantial systematic-error components on top of any shared "difficulty" axis. Distinguishing these is left for follow-up work with a larger and more diverse panel.

H. Reproducibility and runtime

All DRfold2 inferences in this study were run on a single RTX-class GPU (one $\sim$3 min call per target on RTX 5090, $\sim$6 min on the same hardware including the CPU-bound `Optimization.py` step). The full 12-target panel took $\sim$1 hour of pod time for $\sim$\$0.30. All AlphaFold 3 calls were one-click Server submissions (default settings, no MSA tuning, no parameter changes); each takes $\sim$5 minutes server-side and returns a ZIP with five seeds and per-atom pLDDT JSONs.

The full analysis pipeline (Kabsch alignment, RMSD computation, Spearman correlations, Bonferroni correction, figure generation) is deterministic and runs in seconds on a laptop CPU from the released intermediate files. The full source tree, AF3 ZIPs, DRfold2 80-model ensembles, and all generated figures are at https://github.com/Vince-Anduril/rna3d-calibration.

V. Discussion

The headline finding is methodological: for transformer-based structure predictors that emit a candidate ensemble before selection, the 80-model standard deviation acts as a calibrated per-residue uncertainty signal of comparable quality to an explicit trained confidence head. This matters for two practical reasons. First, it gives users of architectures without a confidence head a post-hoc UQ procedure that requires no retraining and no model changes — just access to the unprocessed ensemble. Second, the observation that the two signals are largely orthogonal suggests they can be combined (e.g. both passing or both failing a per-residue cutoff) for stronger selective-prediction guarantees than either signal alone provides.

A second result is that AlphaFold 3's pLDDT calibration, when measured directly against an X-ray crystal of the same molecule, survives Bonferroni correction on half of our blind panel even when its absolute pLDDT is in the "low-confidence" regime. This contradicts the simplifying narrative that low-pLDDT implies an unusable prediction — in our data, low-pLDDT residues are systematically further from the crystal than high-pLDDT residues within the same target, regardless of the absolute pLDDT range.

VI. Limitations

- $N = 12$ targets, all RNA structures of length 57–101 nt. The pattern may not hold at larger sizes or in other biomolecule classes (proteins, complexes).
- AF3 was run through the public Server with default settings; the per-seed predictions are not reproducible bit-for-bit across re-runs because the Server's internal MSA build is not deterministic.
- Bonferroni correction is conservative; the Benjamini–Hochberg false-discovery-rate at $q = 0.05$ would accept additional targets.
- The cross-signal orthogonality test ($\rho_{\text{cross}} = -0.24$, $p \approx 0.05$) is statistically borderline; an out-of-distribution validation on a larger panel would settle whether the two signals are genuinely orthogonal or only weakly anti-correlated.
- No fairness-of-comparison claim is made between AF3 and DRfold2 on absolute prediction quality; that is the subject of a separate companion paper.

A. Future work

Three concrete extensions would tighten this work into a definitive statement about ensemble-disagreement UQ:

1) **Larger and out-of-domain panel.** Extending the panel from $N = 12$ to $N \gtrsim 50$ RNA crystals, ideally drawing on the next post-cutoff batch (deposit dates 2026 or later for both models), would give the cross-signal orthogonality test (§ IV-B) statistical power and would resolve whether the two signals' Bonferroni-survival rates remain at $\sim$50% or converge to a stable fraction. Adding protein structure prediction targets (AlphaFold 3 covers both) would test whether the ensemble-std procedure of Algorithm 1 generalises beyond RNA.
2) **Multi-seed AF3 as an alternative implicit signal.** AF3 itself emits five seeds per Server call. Treating the five-seed C1$'$ spread as an ensemble-disagreement signal analogous to DRfold2's 80-model std would let us compare the implicit vs. explicit signals *within the same model*, which is a cleaner test than the across-model comparison we report.
3) **Conformal calibration of the joint cutoff.** Section IV-C proposes a joint cutoff using both signals; calibrating it with split conformal prediction [9] on a held-out fraction of the panel would convert the empirical Spearman correlations into a calibrated coverage guarantee, which is what a deployed selective-prediction system actually needs.

VII. Conclusion

On a panel of $N = 12$ post-cutoff RNA X-ray crystals, we have shown that DRfold2's 80-model ensemble standard deviation acts as a calibrated per-residue uncertainty signal of quality comparable to AlphaFold 3's explicit pLDDT — $6/12$ Bonferroni-significant targets each, with peak per-target correlations of $\rho = +0.66$ and $\rho = -0.60$ respectively. The two signals are largely orthogonal: they flag different residues as uncertain on the same molecule. The methodological consequence is that transformer-based structure predictors with internal ensembles can support per-residue selective prediction even without an explicit trained confidence head, by reading their unprocessed ensemble post-hoc.

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References

[1] Lee, Y. *et al.* "DRfold2: a deep learning method for RNA tertiary structure prediction using single-sequence input." *PLOS Biology* (2026).

[2] Abramson, J. *et al.* "Accurate structure prediction of biomolecular interactions with AlphaFold 3." *Nature* 630, 493–500 (2024).

[3] Shen, T. *et al.* "Accurate RNA 3D structure prediction using a language model-based deep learning approach." *Nature Methods* 21, 1545–1554 (2024).

[4] Wohlwend, J. *et al.* "Boltz-1: democratising biomolecular interaction prediction." bioRxiv (2024).

[5] Przytula-Mally, A. *et al.* "Anticodon-like loop-mediated dimerization in the crystal structures of HDV-like CPEB3 ribozymes." bioRxiv 2022.09.22.508989 (2022). PDB 7QR3.

[6] El-Yaniv, R., Wiener, Y. "On the foundations of noise-free selective classification." *Journal of Machine Learning Research* 11, 1605–1641 (2010).

[7] Lakshminarayanan, B., Pritzel, A., Blundell, C. "Simple and scalable predictive uncertainty estimation using deep ensembles." *Advances in Neural Information Processing Systems* 30 (2017).

[8] Gal, Y., Ghahramani, Z. "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning." *International Conference on Machine Learning* (2016).

[9] Vovk, V., Gammerman, A., Shafer, G. *Algorithmic Learning in a Random World*. Springer (2005).

[10] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q. "On calibration of modern neural networks." *International Conference on Machine Learning* (2017).

[11] Geifman, Y., El-Yaniv, R. "Selective classification for deep neural networks." *Advances in Neural Information Processing Systems* 30 (2017).

[12] Das, R. *et al.* "Assessment of three-dimensional RNA structure prediction in CASP15." *Proteins* (2023).